

# Erdős Problem #995

EP-995

L1 - TRACTABLE

PARTIALLY SOLVED

Number Theory

## ORIGINAL PROBLEM STATEMENT

Let  $n_1 < n_2 < \dots$  be a lacunary sequence of integers and  $f \in L^2([0, 1])$ . Estimate the growth of, for almost all  $\alpha$ ,  $\sum_{1 \leq k \leq N} f(\{\alpha n_k\})$ . For example, is it true that, for almost all  $\alpha$ ,  $\sum_{1 \leq k \leq N} f(\{\alpha n_k\}) = o(N\sqrt{\log \log N})$ ?

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Archive: <https://www.unsolvedmath.com/problems?difficulty=1>  
Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>